BLWP Waterway Contamination Progress Report 3

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Contributions

Evan Marino: slide 3 on river flow and the corresponding d3 graph with it (https://observablehq.com/d/f3a5990cad5b9d5a)

Dylan Long: Slide 4 - Developed 2 variable D3 bar chart exploring measurements and contamination(measurements_barchart.html)

Dylan Long: Slide 5 - Developed interactive D3 line chart exploring contamination values for each chemical over time in each location(line_plot_measurements.html)

Dylan Long: Slide 9 - Drew conclusion based on provided evidence

Alex Schifferle: Slide 6 - Created animated bar graph to show how harmful diet could be, along with an explanation, as well as mentioning other chemicals.

Alex Schifferle: Slide 9 - Helped with second part of the conclusion and added a visual to display water flow, showing other chemicals as well.

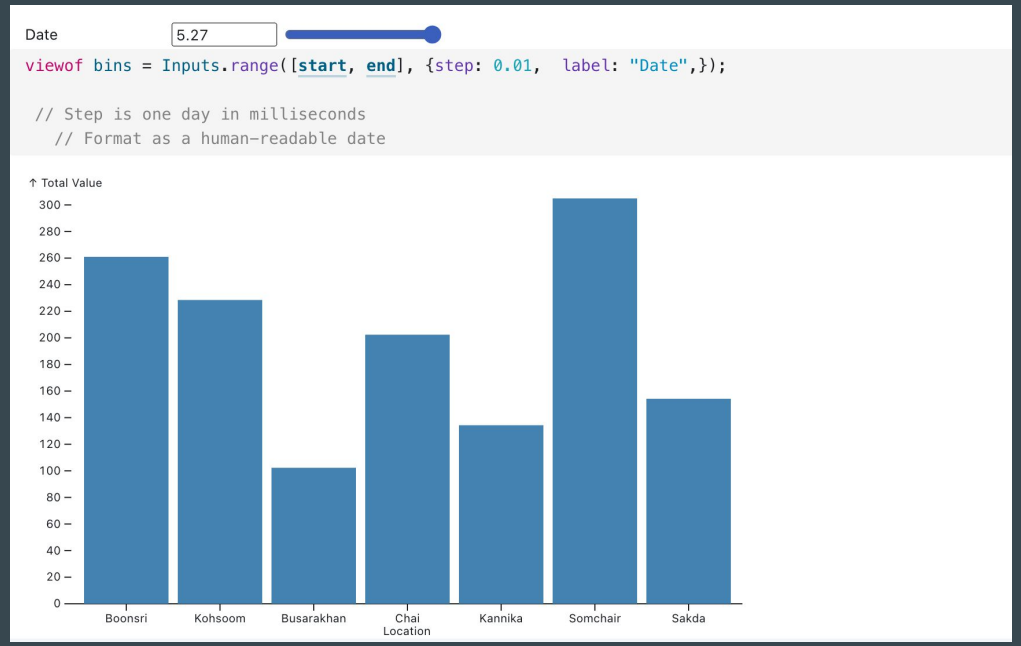
Luke Dasilva: Slide 7: created the live line graph to show the levels of four different chemicals (Ammonium, Nitrites, Nitrates, and Dissolved Oxygen) throughout the years that affect the water the pipits and wildlife drink.

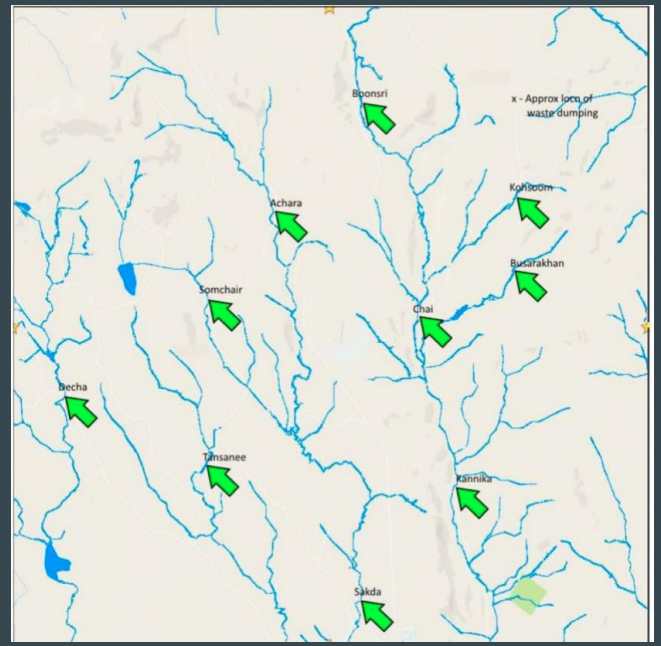
Luke Dasilva: Slide 8: created the live bar graph that shows the same four chemicals in the previous slide, but now in each sampling locations and shows which locations have higher readings of these chemicals.\

Santiago Alvarez (slide 9) : representation of the 7 most toxic elements collected during 1998-2016, shown by the colors.

D3 graphs and data: https://github.com/longdy75/VAST

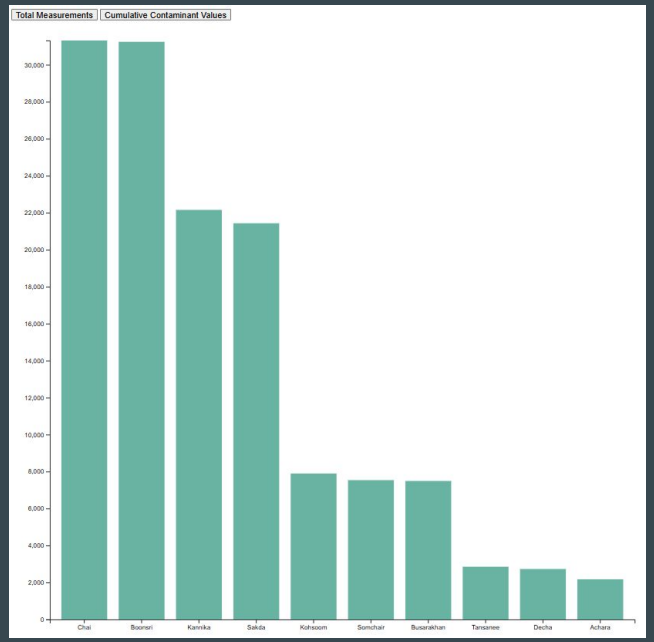
River Flow

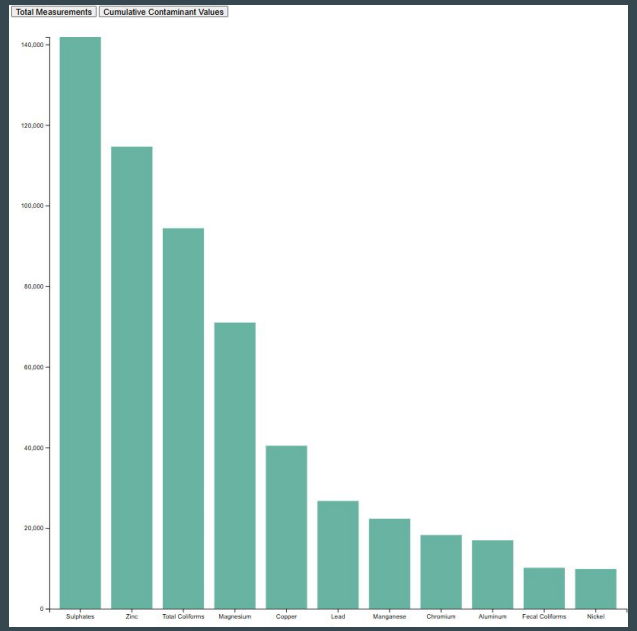




- With the presentation of the d3 graph it shows how the river system is going from the top-down, proving that the water contamination originates at the suspected dump site

D3 Total Measurements & Top Contam. Values





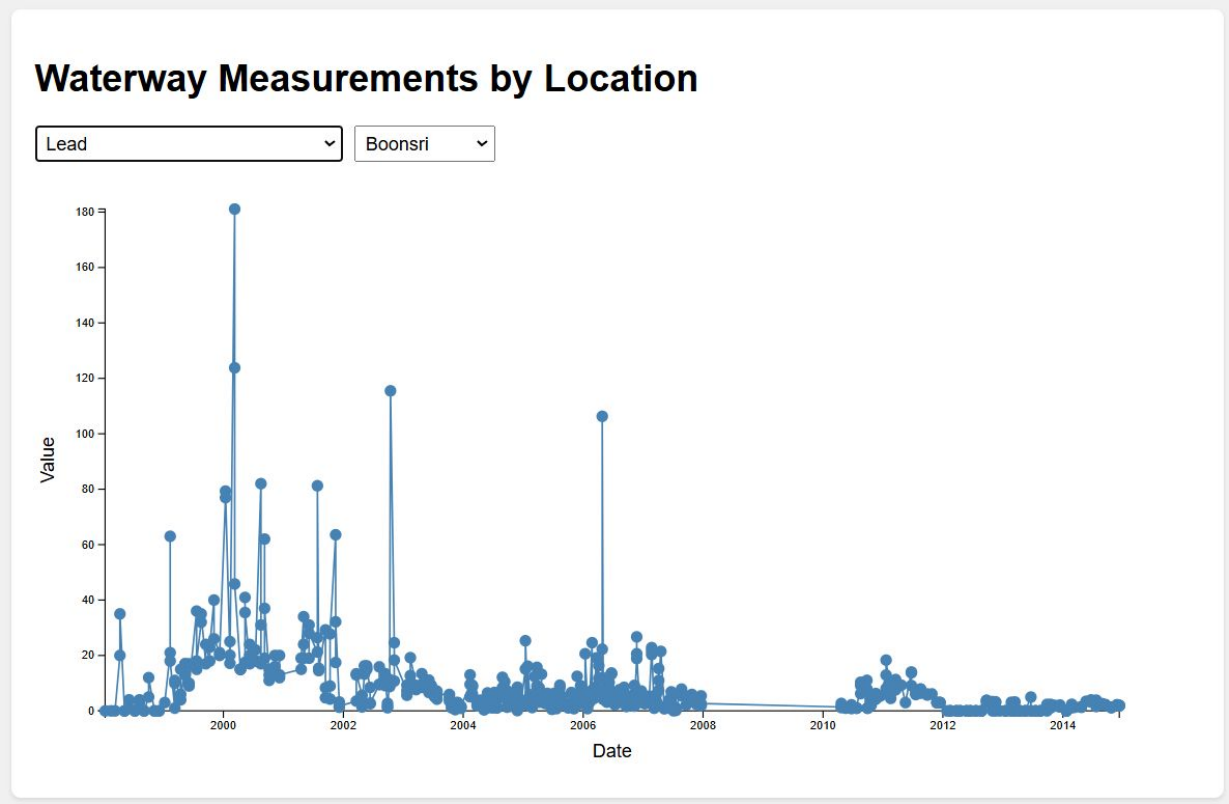
Features:

- 2 Variable Analysis
- Scaling Axes
- Animated Transitions

Findings:

- Sites were measured very inconsistently
- Considerable levels of heavy metals found

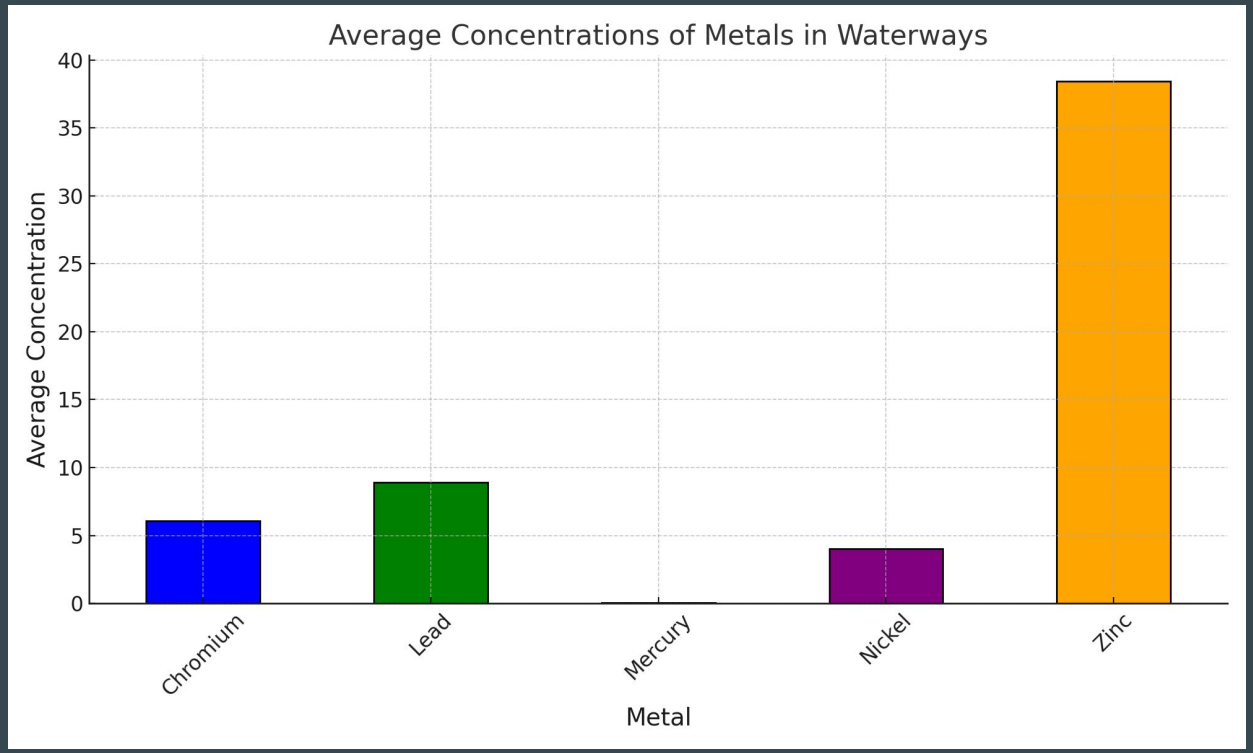
D3 Measurements by Location



Features:

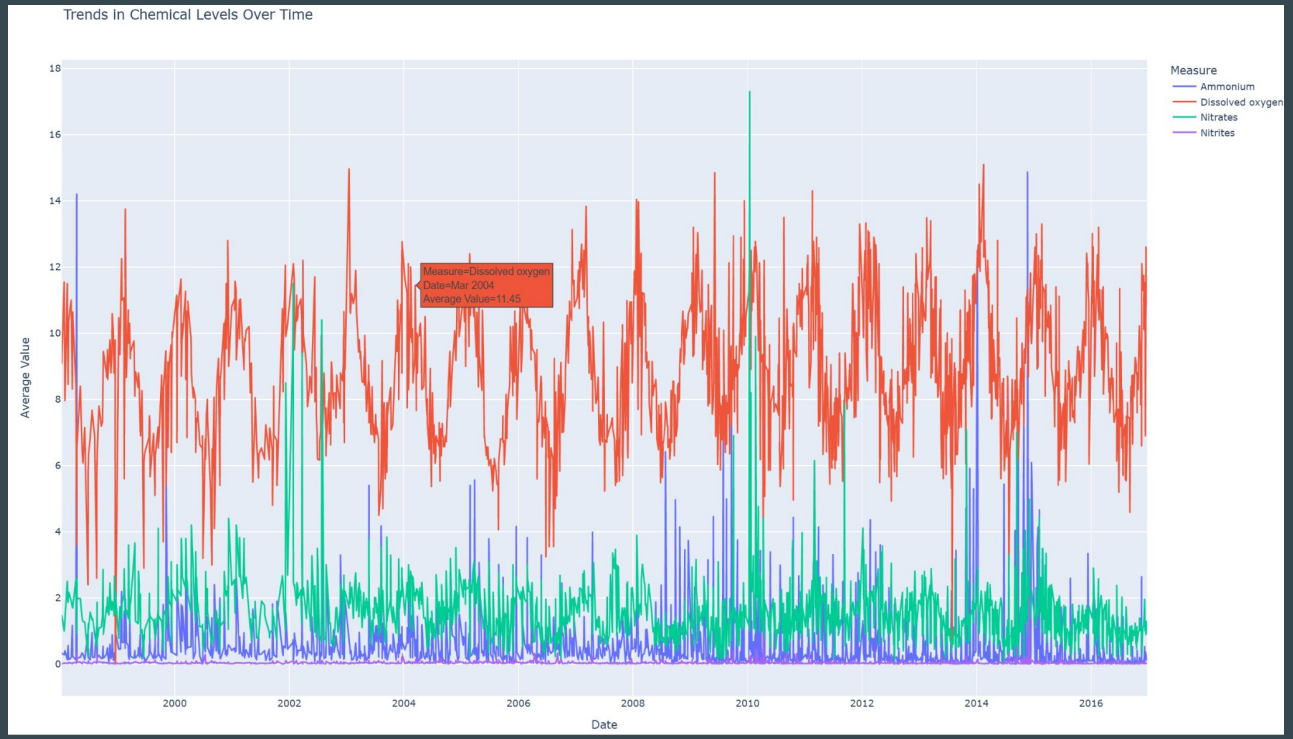
- Double dropdown menu
- Tooltip showing specific date & value
- Dots to show individual points
- Connecting lines to show movement
- Scaling axes
- Animated transitions

Average Levels of Chemicals within Diet



- Specifically these cling to the tissue of fish.
- Pipits can potentially eat the fish.
- This can vary with age
- There are other elements besides these just in smaller but not insignificant amounts. (They also don't necessarily impact diet)

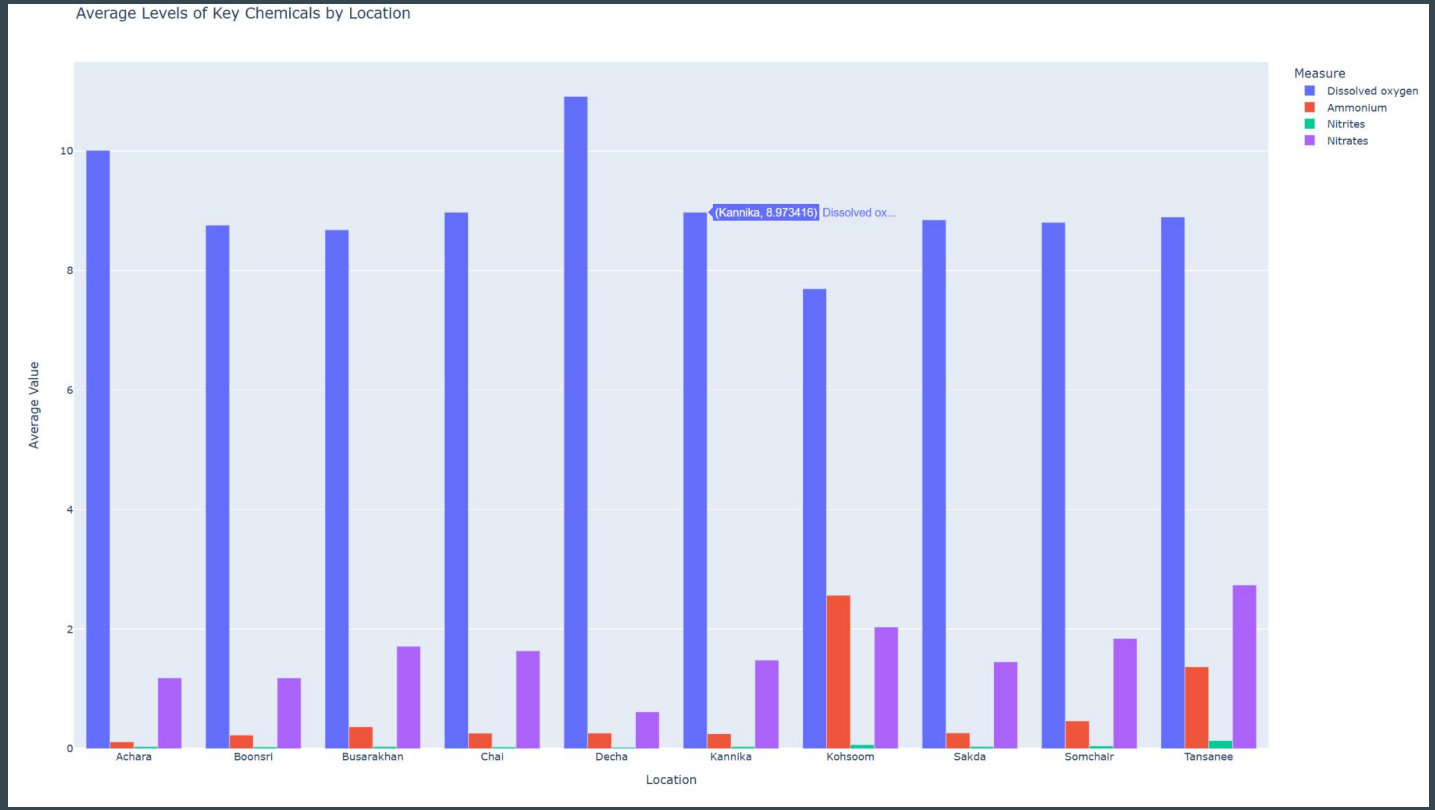
Chemicals That Affect Pipits and Other Wildlife:



This Line Graph shows the average values of these four key chemicals (Ammonium, Nitrites, Nitrates, and Dissolved Oxygen) over time, showing an increase or decrease in patterns.

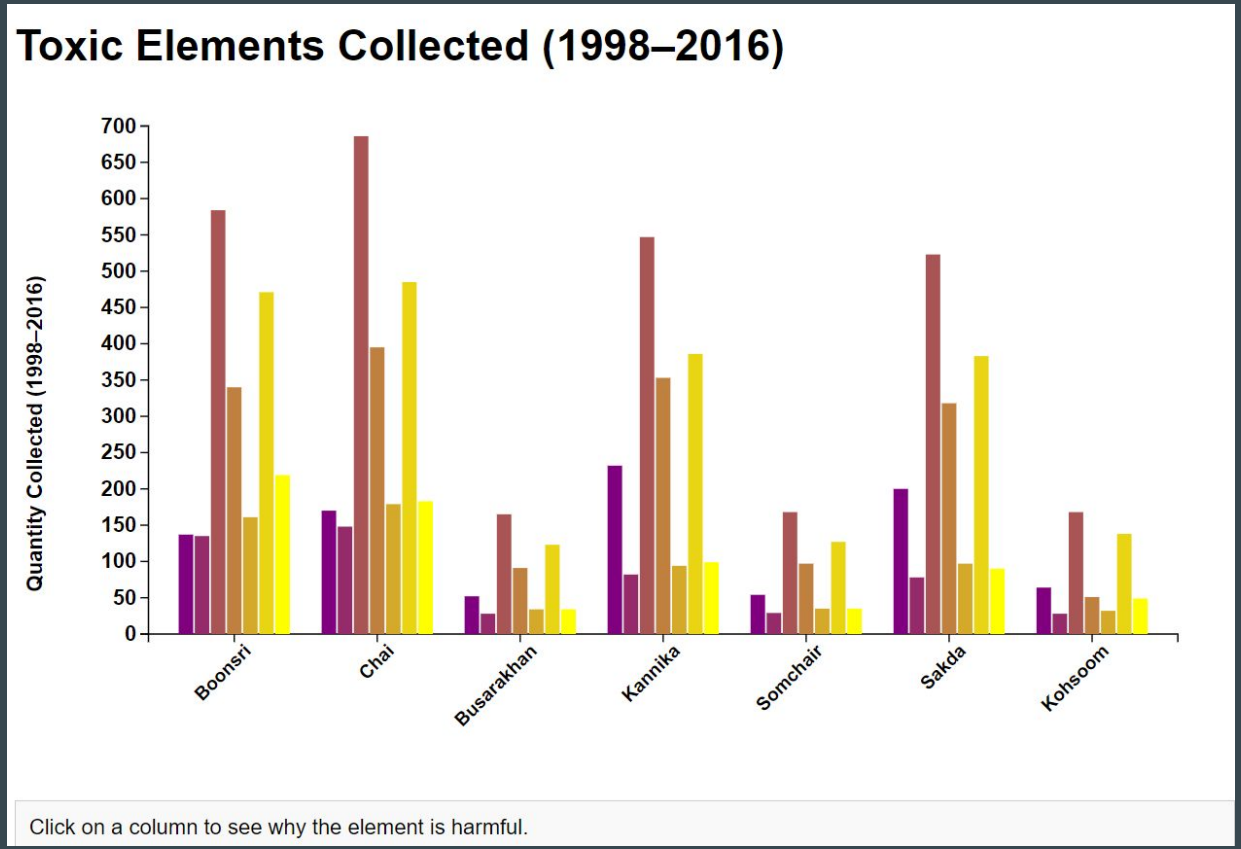
- Can move cursor throughout graph to see the exact date measured and the average value
- Color coded for each chemical
- Starts in 1998 till 2016 in 2 year intervals labeled

Chemicals That Affect Pipits and Other Wildlife:



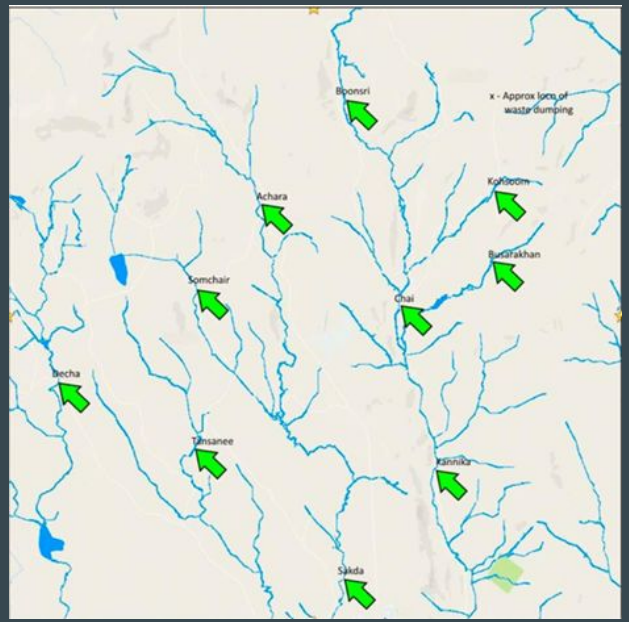
This Bar chart compares the average levels of the four chemicals across the different sampling locations

- Can hover over each bar and see the location, and average value of each of the four chemicals.
- Color coded just like the previous graph for each chemical.
- Shows Dissolved Oxygen has the highest reading in each location



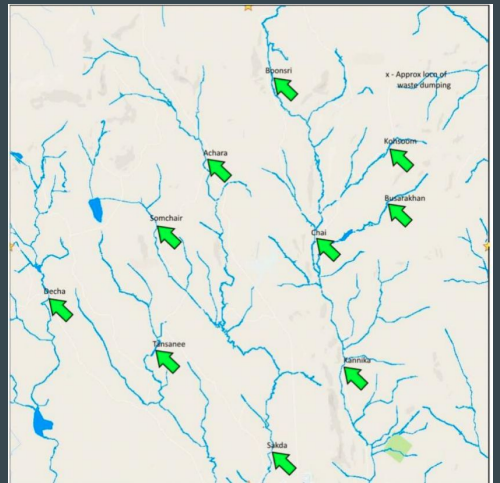
Elements from left to right :

Arsenic Alachlor Cadmium
Mercury P,P-DDD P,P-DDT
Endrin



Conclusion

- Kasios Furniture Company is responsible for contaminating the Mistford preserve
  - **Evidence:**
    - Chemical readings traveling down river from the suspected dumping location (See slide 3)
    - Excessive levels of heavy metals:
      - Lead
      - Chromium
      - Nickel
      - Mercury
        - Cadmium, Arsenic, Alachlor, Fluoranthene, Heptachlor, Endrin

As shown here you can see that the rivers flowing down are closest to the waste and they also contain the highest concentrations. This not only affects the water but it can have in impact on other surrounding wildlife, which can also have an impact on diet and various other ways of life..

(See slide 3 for more detail of the chemicals)